Jose Raul Rincon Garcia

Rice University
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Houston, Texas, 77005

Research Assistant
Department of Civil and Environmental Engineering
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EDUCATION

Rice University, PhD in Civil Engineering — Houston, TX, USA
Research topic: "Smart and Equitable Resilience Measuring and Modeling for Dynamic Infrastructure Systems". *Advisor*: Jamie E. Padgett, PhD. — Degree expected 2025

Universidad de los Andes, Master of Science in Civil Engineering — Bogotá, Colombia
Thesis: "Seismic vulnerability assessment of reinforced concrete buildings using non-linear dynamic analysis". *Advisor*: Luis E. Yamin, PhD. — July, 2013 – June, 2015

Universidad de los Andes, Bachelor of Science in Civil Engineering — Bogotá, Colombia
Thesis: "Design's guide for buildings with Steel Plate Shear Walls". *Advisor*: Juan C. Reyes, PhD. — July, 2009 – June, 2013

GRANTS AND AWARDS

2023 Robert P. and Eleanor Warden Shubinski Awardee, Rice University — Spring, 2023

2021 H. W. Reeves Endowed Scholarship Awardee, Rice University — Spring, 2021

Outstanding Journal Paper Award for 2018, — January, 2020
ASCE Journal of Performance of Constructed Facilities.

Fulbright Scholar, Fulbright-Minciencias (Colombia). — August, 2019

Innovative Teacher Recognition, at Universidad de los Andes, Colombia. — June 20th, 2019

Mario Galan Gomez Scholarship, granted by ECOPETROL, Colombia. — 2009 – 2013

RESEARCH EXPERIENCE

Rice University — Houston, TX, USA
Research Assistant (PI: Jamie E. Padgett, PhD) — 2020 – Actual

- Develop resilience modeling approaches, for multi-scale infrastructure systems, that consider smart strategies to analyze low-probability events, compute metrics in multi-scale and heterogeneous systems, and cope with ever-changing conditions (such as aging, deterioration, or technology shifts).

- Propose methods to improve the science of measuring, comparing, and updating infrastructure systems performance to attain accurate, unbiased, and inequitable estimates.
- Research on multi-scale uncertainty propagation aided by machine learning and statistical learning techniques for instant and long-term assessment of civil infrastructures, and continual learning.
- Develop Usecases for the DesignSafe Cyberinfrastructure; evaluate interoperability between DesignSafe and INCORE through Jupyter Notebooks.
- Perform large-scale simulations using the high-performance computing capacities at the Texas Advanced Computer Center (TACC).

Universidad de los Andes Bogotá, Colombia
Research Assistant (PI: Luis E. Yamin, PhD) 2015 – 2017

- Participated in the development of a probabilistic methodology to include the soil response-related uncertainties in microzonation studies; the case study was the Aburrá Valley (Colombia)
- Coordinated the development of a large-scale seismic risk buildings portfolio model for the Metropolitan Area of the Aburrá Valley to predict probable losses; stress on vulnerability assessment of masonry houses
- Contributed with faculty and graduate researcher to develop an exposure model of the Bogota's buildings for the risk assessment in the project SATREPS, directed by The Japan International Cooperation Agency-JICA and the Japan Science and Technology Agency -JST
- Evaluated and developed, as part of a consultant team, the national-scale seismic risk mitigation strategy in the Peruvian school infrastructure

Universidad de los Andes Bogotá, Colombia
Graduate Research Student (PI: Luis E. Yamin, PhD) 2013 – 2015

- Researched and collaborated in the development of a probabilistic methodology for seismic vulnerability assessment of buildings using part of the FEMA P-58 project results
- Collaborated in the site-investigations, field and laboratory tests and investigation of gathered information in a research case study of a collapsed 27-story building in Medellin, Colombia
- Built the computational models for structural analysis and seismic assessment and the revision of the code-compliance (focused in structural design requirements) for residential buildings in Medellin and Bogota
- Planed, coordinated, measured and processed the signals of noise and ground vibration produced by the blasting operations in a carbon mine in Colombia in order to identify its impact on nearest communities and the local constructions

PUBLICATIONS

Journal Articles Published

1. Rincon R, Padgett JE. (2023) "Fragility Modeling Practices and their Implications on Risk and Resilience Analysis: from the Structure to the Network Scale" *Earthquake Spectra,* Prepublished on January 2, 2024. https://journals.sagepub.com/doi/abs/10.1177/87552930231219220.

2. Miura H, Matsuoka M, Reyes JC, Pulido N, Hashimoto M, Riaño AC, Hurtado A, Rincon R, García H, Lozano C. Quick Estimation Model for Mapping Earthquake Impacts in Bogotá, Colombia. *ISPRS International Journal of Geo-Information*. 2023; 12(12):471. https://doi.org/10.3390/ijgi12120471.

3. Rincon R, Reyes JC, Carrillo J, Clavijo-Tocasuchyl A. (2022) "Empirical Fragility Assessment of Adobe and Rammed Earth Walls Subjected to Seismic Actions." *Earthquake Engineering & Structural Dynamics*, 51, no. 5 (2022): 1133–57. https://doi.org/10.1002/eqe.3608.

4. Buitrago-Flórez F, Reyes JC, Rincon R, Hernández C, Galvis FA, Ángel C. (2020). Engaging in homework development: TARSIS platform as an innovative learning methodology. *Australasian Journal of Educational Technology*, 36(3), 147-162. https://doi.org/10.14742/ajet.5865

5. Reyes JC, Rincon R, Yamin LE, Correal JF, Martínez JG, Sandoval JD, Gonzalez CD, Ángel CC. (2019). Seismic retrofitting of existing earthen structures using steel plates. *Construction and Building Materials*, 230, 117039. Elsevier. https://doi.org/10.1016/j.conbuildmat.2019.117039

6. Yamin LE, Reyes JC, Rueda R, Prada E, Rincon R, Herrera C, Daza J, Riaño AC. (2018). Practical seismic microzonation in complex geological environments. *Soil Dynamics and Earthquake Engineering*, 114, 480–494. Elsevier. https://doi.org/10.1016/j.soildyn.2018.07.030

7. Medina JM, Villate C, Caicedo B, Yamin LE, Correal JF, Estrada N, Rincon R. (2018). Riesgo sísmico, geotécnico y estructural. El caso de estudio de los edificios San Jerónimo de Yuste en Bogotá. *Informes de la Construcción*, 70(550): e251. https://doi.org/10.3989/ic.16.131 (in Spanish)

8. Yamin LE, Correal JF, Reyes JC, Ramírez F, Rincon R, Hurtado AI, Dorado JF. (2018). Sudden Collapse of the 27-Story Space Building in Medellin, Colombia. *Journal of Performance of Constructed Facilities*, 32(3), 1–13. https://doi.org/10.1061/(ASCE)CF.1943-5509.0001148

9. Yamin LE, Hurtado AI, Rincon R, Dorado JF, Reyes JC. (2017). Probabilistic seismic vulnerability assessment of buildings in terms of economic losses. *Engineering Structures*, 138, 308–323. Elsevier. https://doi.org/10.1016/j.engstruct.2017.02.013

Journal Articles Accepted

10. Makhoul N, Roohi M, van de Lindt JW, Sousa H, Santos LO, Argyroudis S, Barbosa A, Derras B, Gardoni P, Lee JS, Mitoulis S, Moffett B, Navarro C, Padgett JE, Rincon R, Schmidt F, Shaban N, Stefanidou S, Tubaldi E, Xenidis Y, Zmigrodzki S. (2023) "Seismic resilience modelling of an interdependent built environment for integrating SHM and emerging technologies in decision-making" *Structural Engineering International,* ***Accepted,*** June, 2023.

Journal Articles in Review

11. Padgett JE, Rincon R, Panakkal P. (2024) "Future cities demand smart and equitable infrastructure resilience modeling perspectives" *Nature Communications,* ***in review,*** January, 2024.

Conference Papers (* denotes presenter)

1. Rincon R*, Padgett JE. (2023). "Smart resilience: capturing dynamic, uncertain and evolving lifecycle conditions". *8th International Symposium on Life-Cycle Civil Engineering (IALCCE 2023)*, Milan, Italy, July 2-6, 2023. In F. Biondini & D.M. Frangopol, Life-Cycle of Structures and Infrastructure Systems (1st ed., pp. 341–348). CRC Press. https://doi.org/10.1201/9781003323020-39.
2. Rincon R*, Padgett JE. (2022) "Seismic reliability analysis of complex structural systems using system-level surrogate models". *Proceedings of the 12th National Conference in Earthquake Engineering*, Earthquake Engineering Research Institute, Salt Lake City, UT, June 27-July 1, 2022.
3. Rincon R, Reyes JC, Carrillo J, Clavijo-Tocasuchyl A*. (2022) "Towards fragility surfaces and data-driven capacity curves for earthen walls". *10th Colombian National Conference on Earthquake Engineering*. Bogotá, Colombia, May 29th to May 31st (in Spanish).
4. Castaño M*, Correal JF, Rincon R. (2022) "Estudio de fragilidad de colapso de pórticos resistentes a momento en estructuras industriales prefabricadas de gran tamaño". *10th Colombian National Conference on Earthquake Engineering*. Bogotá, Colombia, May 29th to May 31st (in Spanish).
5. Reyes JC*, Yamin LE, Rueda R, Prada E, Rincon R, Herrera C, Daza J, Riaño AC. (2019). "Seismic microzonation of the Medellin area using a probabilistic approach". *VII International Conference on Earthquake Geotechnical Engineering*. Rome, Italy. June 2019.
6. Rincon R, Yamin LE, Arboleda C, Oliveros C*, García AP, Galvis F. (2019) "Repair/Replacement Costs Evaluation of Post-Earthquake Affected Bridges Methodology". *9th Colombian National Conference on Earthquake Engineering*. Cali, Colombia, May 29th to May 31th (in Spanish).
7. Yamin LE, Reyes JC, Echeverry J*, Correal J, Ramirez F, Rincon R, Galvis F, Prada E. (2019) "Seismic risk mitigation strategies for urban bridges". *9th Colombian National Conference on Earthquake Engineering*. Cali, Colombia, May 29th to May 31st (in Spanish).
8. Rueda M*, Rincon R, Yamin LE. (2019) "Assessment of the Seismic Response of Reinforced Concrete Moment Frames as Function of the Intensity Measure". *9th Colombian National Conference on Earthquake Engineering*. Cali, Colombia, May 29th to May 31st (in Spanish).
9. Spinel J*, Reyes JC, Rincon R, Yamin LE. (2019) "Estimation of seismic demand parameters in existing unreinforced masonry housing units". *9th Colombian National Conference on Earthquake Engineering*. Cali, Colombia, May 29th to May 31st (in Spanish).
10. Fernández R*, Rincon R, Yamin LE. (2019) "Uncertainty in the Obtained Benefits for Different Retrofitting Alternatives". *9th Colombian National Conference on Earthquake Engineering*. Cali, Colombia, May 29th to May 31st (in Spanish).
11. Aguirre L*, Rincon R. (2019) "Correlation between damage states of elements and their impact on seismic vulnerability". *9th Colombian National Conference on Earthquake Engineering*. Cali, Colombia, May 29th to May 31st (in Spanish).
12. Yamin LE, Rincon R*, Fernández RI, García AP, Reyes JC. (2018) "Repair Cost and Downtime Seismic Vulnerability Assessment for RC School Buildings". *11th U.S. National Conference on Earthquake Engineering. Integrating Science, Engineering and Policy*. Los Angeles CA, USA, June 25-29, 2018.
13. García A*, Yamin LE, Rincon R. (2017) "Topographic amplification effects in slope zones and its integration in seismic microzonation studies". *8th Colombian National Conference on Earthquake Engineering. Bogotá, Colombia, May 31-June 2, 2017.*

14. Prada E*, Yamin LE, Rueda R, Rincon R. (2017) "Uncertainty in the one-dimensional dynamic response of soil reservoirs for seismic microzonation effects". *8th Colombian National Conference on Earthquake Engineering. Bogotá, Colombia, May 31-June 2, 2017.*
15. Aroquipa H*, Rincon R, Fernández R, Yamin LE. (2017) "Evaluation of incremental seismic retrofitting techniques for typical school buildings in Peru". *8th Colombian National Conference on Earthquake Engineering. Bogotá, Colombia, May 31-June 2, 2017.*
16. Rincon R*, Yamin LE, Becerra AF. (2017) "Seismic risk assessment of public schools and prioritization strategy for risk mitigation". *16th World Conference on Earthquake Engineering*. Santiago de Chile, Chile, January 9th to 13th, 2017.
17. Yamin LE*, Hurtado AI, Rincon R. (2017) "Controlling parameters in the assessment of the seismic vulnerability of buildings". *16th World Conference on Earthquake Engineering*. Santiago de Chile, Chile, January 9th to 13th, 2017.
18. Rincon R*, Yamin LE, Hurtado AI, Pulido JF. (2015) "Seismic evaluation and performance of non-conventional structures: Bulk Storage Structures" (in Spanish). *7th Colombian National Conference on Earthquake Engineering*. Bogotá, Colombia, May 2015.
19. Hurtado A*, Yamin LE, Rincon R, Pulido JF, Dorado JF, Reyes JC. (2015) "Incremental dynamic analysis, a tool for seismic vulnerability assessment of buildings" (in Spanish). *7th Colombian National Conference on Earthquake Engineering*. Bogotá, Colombia, May 2015.
20. Dorado JF*, Yamin LE, Hurtado AI, Rincon R, Torres ID. (2015) "Cost model of structural and non-structural elements for the evaluation of the seismic vulnerability of reinforced concrete buildings" (in Spanish). *7th Colombian National Conference on Earthquake Engineering*. Bogotá, Colombia, May 2015.
21. Pulido JF*, Yamin LE, Rincon R. (2015) "Seismic risk assessment of the main cities of Colombia" (in Spanish). *7th Colombian National Conference on Earthquake Engineering*. Bogotá, Colombia, May 2015.
22. Yamin LE, Hurtado AI, Rincon R, Barbat AH, Reyes JC. (2014) "Use of non-linear dynamic analysis in the assessment of seismic vulnerability of buildings". *2nd European Conference on Earthquake Engineering and Seismology*. Istanbul, Turkey, August 2014.
23. Yamin LE, Hurtado AI, Rincon R, Pulido JF, Reyes JC, Barbat AH. (2014) "Evaluation of seismic code specifications using static nonlinear analyses of archetype buildings". *10th U.S. National Conference on Earthquake Engineering. Anchorage*, Alaska, USA, July 2014. Poster presentation delivered.

Book Chapters and Book Sections

1. Rincon R, Camacho R, Zuluaga S, Avalo A, Ramírez A. (2021) Section 2, Study Case N° 4. "Statics, a case of cross-sectional design of the evaluation system by Big Ideas theory" (Galvis A, Editor) *In Course design by Big Ideas, using active pedagogy and digital technologies.* In Spanish. (Ebook, ISBN: 978-958-798-039-4). Ediciones Uniandes-Universidad de los Andes, 2021.

Datasets, algorithms, and others

1. Rincon R, Padgett, JE. (2023) "Jupyter notebook for visualization of spatially distributed data in risk and resilience analysis." *DesignSafe-CI*. https://doi.org/10.17603/ds2-a74m-g031 v1.

LECTURES, SEMINARS, AND OTHER PRESENTATIONS

Lectures, Seminars, Workshops, and Invited Talks (* denotes presenters)

1. Padgett JE*, Rincon R*, Del-Castillo-Negrete C.*. (2023). "Part II: Supporting Infrastructure Risk and Resilience Research through NHERI DesignSafe Resources," *Workshop: IN-CORE and DesignSafe at ASCE INSPIRE Conference 2023*, Arlington, VA, November 15.
2. Padgett JE*, Rincon R. (2022). "Facilitating seismic risk and resilience analysis in heterogenous and multi-scale systems", *4th Kenji Ishihara Colloquium Series on Earthquake Engineering: "Artificial Intelligence Applications in Earthquake Engineering"*, San Diego, CA, September 1-2.
3. Reyes JC*, Rincon R*. (2019). "Innovation in the structural systems analysis course for civil engineers" (in Spanish). *Workshop on New Alternatives in Engineering Education*. School of Engineering, Universidad de los Andes, Bogotá, Colombia. June 17th 2019.
4. Reyes JC*, Rincon R*. (2019). "Actively involve students in the development of assignments using the TARSIS platform" (in Spanish). *Cycle of Webinars - Exchange of experiences in blended and virtual courses*. Academic Vice-Presidency, Universidad de los Andes, Bogotá, Colombia. July 17th 2019.
5. Rincon R*, Camacho R*. (2018). "Innovation processes for the Statics course supported by Conecta-Te". *Dialogues on participation in innovation with Conecta-TE*. Department of Civil and Environmental Engineering, Universidad de los Andes. Bogotá, Colombia. October 26th 2018.
6. Camacho R*, Rincon R*. (2018). "Innovation in the course of Statics supported by Information and Communication Technologies". *Dialogues on participation in innovation with Conecta-TE*. Department of Mechanical Engineering, Universidad de los Andes. Bogotá, Colombia. October 5th, 2018.

Conferences, Symposiums and Poster presentations (* denotes presenters)

7. Rincon R*, Padgett JE. (2023). "Exploration of biasedness and inequities in infrastructure resilience modeling" Exploration of biasedness and inequities in infrastructure resilience modeling, Arlington, VA, November 16-18.
8. Rincon R*, Padgett JE (2022). "Traditional and surrogate-based procedures for fragility functions derivation and their influence on network analysis", *2022 SimCenter Symposium*. Austin, TX, November 3-4.
9. Reyes JC*, Rincon R, Yamin LE, Correal J, Echeverry J. (2019). "Behavior of adobe and rammed earth buildings and possible options for vulnerability mitigation". *Jornadas XVIII Geotécnicas y XXI Estructurales*. Bucaramanga, Colombia. Nov 2019.
10. Rincon R*, Camacho R*. (2017). " Innovaciones en el curso de estática usando TIC". *Innovación en la educación con Tecnologías de la Información y Comunicación (TIC)*. Facultad de Educación y Centro de Innovación en Tecnología y Educación, Universidad de los Andes. Bogotá, Colombia. Poster presentation. November 16th, 2017.

TEACHING EXPERIENCE

Rice University Houston, TX, USA

Co-leading the Resilience Seminar of the Gulf Scholars Program at Rice University (GSP@RICE) Fall 2022

- Developed the syllabus and co-organized the seminar sessions, co-curricular activities and the assignments for the seminar for the cohort '22.

- Co-lead the seminar sessions and engaged the students to find their future pathways extensions in the GSP@RICE and as change-agents.

Universidad de los Andes Bogotá, Colombia
Full time Lecturer and Researcher (School of Engineering) 2017 – 2020

- Lectured undergraduate courses:

  Structural Design: Leaded a total course re-organization by updating lectures and exams format focused on student-centered approaches. Collaborated in the development of a "Teaching notes book" — Aug. 2019 – Jul. 2020

  Mechanics of Materials: Updated and developed training material to improve students understanding. — Jan. 2018 – Jul. 2020

  Structural Analysis: Contributed to the development *TARSIS* platform, an online resource for student's assignments which uses the student's errors as a tool for self-learning. — Aug. 2017 – Jul. 2020

  Statics: Implemented flipped classroom methodologies as a tool for better understanding supported by the university's group for pedagogical innovation — Aug. 2017 – Aug. 2019

- Guided undergraduate (senior) research projects (required for bachelor's degree)
- Developed course contents, grading course performance and holding regular office hours.
- Contributed in research projects related to risk analysis on critical infrastructure such as bridges portfolios, buildings or educative infrastructure to develop action plans for risk mitigation
- Participated at the *Global Program for Safer Schools,* funded by GFDRR.
- Motivated and investigated the implementation of innovative pedagogical strategies for student-centered learning approaches and the active engagement of the students.
- Lectured undergraduate courses:

STUDENTS GUIDANCE

Undergraduate students at Universidad de los Andes

- Laura Cristina Aguirre García (2018)
  Research Project Advisor. "Components' damage correlation and its impact to vulnerability".
- María Paula Delgado Enríquez (2018)
  Research Project Advisor. "Seismic community resilience: preliminary evaluation of two existing methodologies applied to school buildings".
- Karen Lorena Domínguez Tarazona (2018)
  Research Project Advisor. "Nonlinear dynamic analysis of isolated buildings".
- Juan Sebastián López Bello (2018)
  Research Project Advisor. "Design of isolated buildings with concentrically braced steel frames".

- Nicolás Parra Montaño (2018)
  Research Project Advisor. "Design of isolated buildings with reinforced concrete frames".
- Daniel Castrillón Campo and Astrid Natalia Gutiérrez Bernal (2019)
  Research Project Advisor. "Tool for approximation of bridge's super and substructure material quantities".
- Angie Paola Oviedo Castaño and Luis Felipe Huertas Medina (2019)
  Research Project Advisor. "Preliminary investigation of new seismic design paradigms in Colombia: structural design using uniform collapse risk".
- María Alejandra Ortiz Bernal (2019)
  Research Project Advisor. "Reinforced concrete frames subjected to tsunami loads".
- Juan Sebastián Parra Tarache (2019)
  Research Project Advisor. "Indirect losses in a simplified bridge network under seismic events".
- María Claudia González Clavijo (2019)
  Research Project Advisor. "Impact of existing approximate fundamental vibration period equations in the nonlinear dynamic analysis of combined systems.
- Andrés Felipe Jiménez Caro (2019)
  Research Project Advisor. "Linear performance analysis of tall buildings under seismic hazards".
- Gustavo Turriago Méndez (2019)
  Research Project Advisor. "Preliminary fragility function for adobe and rammed earth walls".
- Cristian Yesid Bula Peña (2020)
  Research Project Advisor. "Non-linear performance analysis of tall buildings under seismic hazards".
- Juan Camilo Vanegas Cañon (2020)
  Research Project Advisor. "Variation in design outcomes of tall buildings given the different existing approximate fundamental vibration period equations".
- Alejandra Clavijo Tocasuchyl (2020)
  Research Project Advisor. "Analysis of collapse risk considering the uncertainties in the fundamental vibration period".

PROFESSIONAL SERVICE

Reviewer for Indexed Journals
Journal of Building Engineering
Journal of Performance of Constructed Facilities

Professional Technical Committees
ASCE Civil Infrastructure and Lifeline Systems Committee (member since 07/2021)
ASCE Risk and Resilience Measurements Committee (member since 07/2021)

Additional Professional Memberships and Activities
Earthquake Engineering Research Institute (EERI), Member
American Society of Civil Engineers, Member
Colombian Association of Earthquake Engineering (Colombia), Member

UNIVERSITY SERVICE AND CONTRIBUTIONS

University Service

- Member of the Strategic Planning Committee for articulation of Rice's vision for the next 10 years (sole representative of graduate students 2023)

Participation in Student Clubs as a Graduate Student at Rice University
Participated in the following clubs (period mentioned in parenthesis):
President, Earthquake Engineering Research Institute – Rice University (2021 – present)
President, Civil and Environmental Engineering Graduate Student Association (2023 – present)
Officer, Civil and Environmental Engineering Graduate Student Association (2021 – 2023)
Officer, Colombian Owls Student Association (2023 – present)
Officer, Latin American Graduate Student Association (2022 – 2023)
Leader of 'The Fulbright Ring' - Fulbright at Rice (2022 – 2023)
Departmental Liaison Center for Teaching Excellence Graduate Advisory Board (2022 – 2023)

CONSULTING & PROFESSIONAL EXPERIENCE

ITEC- Ingeniería Técnica y Científica Bogotá, Colombia
Project specialist 2013 - 2017

- Evaluation of Ecuador's seismic risk profile at the country level. (Client: Inter-American Development Bank - IDB).
- Technical assistance to the World Bank and the Government of Ecuador in structural and construction engineering in support of the response to the April 16th, 2016 Ecuador earthquake (Client: World Bank).
- Exceedance loss curve for frequent events in Honduras (Client: Inter-American Development Bank - IDB).
- Condition and safety level of the foundation of the 15,000 ton cement silo (Client: HOLCIM-Colombia)
- Condition and safety level of the 20,000 ton Clinker Silo (Client: HOLCIM-Colombia)
- Assessment of the disaster risk in Colombia, as part of the risk financing and insurance program (Client: World Bank).

CIVECO S.A.S Bogotá, Colombia
Co-Founder and Project specialist 2016-2017

- Consultancy projects focused in the vulnerability assessment of structures
- Design of residential and commercial low and mid-rise buildings